JEE Mains 2019 Chapter wise Question Bank

Alternating Current - Questions

Q1

A series AC circuit containing an inductor (20 mH), a capacitor (120 μF) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy dissipated in the circuit in 60 s is:

(1) 5.65×10^2 J (2) 2.26×10^3 J
(3) 5.17×10^2 J (4) 3.39×10^3 J

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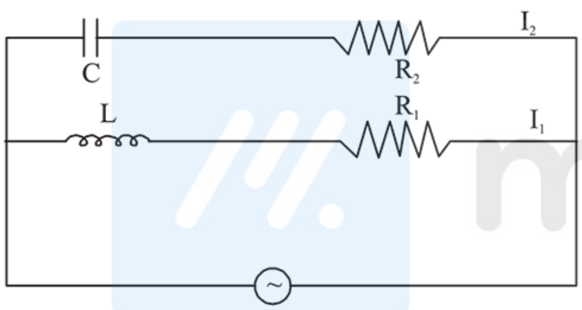
Q2



In the above circuit, $C = \frac{\sqrt{3}}{2}\mu F$, $R_2 = 20\,\Omega$, $L = \frac{\sqrt{3}}{10}$ H and $R_1 = 10\,\Omega$. Current in L-R_1 path is I_1 and in C-R_2 path it is I_2. The voltage of A.C source is given by, $V = 200\sqrt{2}\sin(100\,t)$ volts. The phase difference between I_1 and I_2 is :

(1) 60° (2) 30°
(3) 90° (4) 0

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Q3

An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50Ω. The time taken for the current to rise from half of the peak value to the peak value is :

(1) 5 ms (2) 2.2 ms (3) 7.2 ms (4) 3.3 ms

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Q4

A circuit connected to an *ac* source of *emf* $e = e_0 \sin(100t)$ with t in seconds, gives a phase difference of $\frac{\pi}{4}$ between the *emf* e and current i. Which of the following circuits will exhibit this ?

(1) RL circuit with R = 1 kΩ and L = 10 mH

(2) RL circuit with R = 1 kΩ and L = 1 mH

(3) RC circuit with R = 1 kΩ and C = 1 μF

(4) RC circuit with R = 1 kΩ and C = 10 μF.

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Alternating Current - Answers

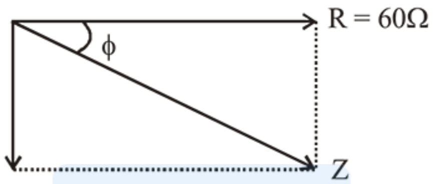
Q1

(3) Given: $R = 60\Omega$, $f = 50$ Hz, $\omega = 2\pi f = 100\pi$ and $v = 24v$
$C = 120\ \mu f = 120 \times 10^{-6} f$

$$x_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 120 \times 10^{-6}} = 26.52\Omega$$

$$x_L = \omega L = 100\pi \times 20 \times 10^{-3} = 2\pi\Omega$$

$$x_C - x_L = 20.24 \approx 20$$



$$z = \sqrt{R^2 + (x_C - x_L)^2}$$

$$z = 20\sqrt{10}\Omega$$

$$\cos\phi = \frac{R}{z} = \frac{60}{20\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$P_{avg} = VI\cos\phi, I = \frac{v}{z} = \frac{v^2}{z}\cos\phi = 8.64 \text{ watt}$$

Energy dissipated (Q) in time $t = 60$s is
$Q = P.t = 8.64 \times 60 = 5.17 \times 10^2$ J

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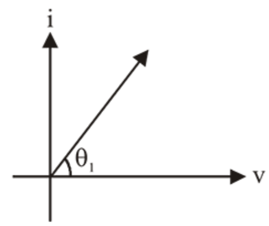
Q2

(Bonus)



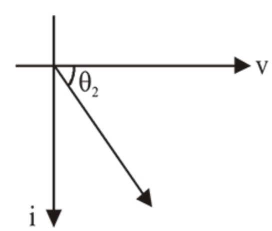
Capacitive reactance,

$$X_c = \frac{1}{\omega C} = \frac{4}{10^{-6} \times \sqrt{3} \times 100} = \frac{2 \times 10^4}{\sqrt{3}}$$

$$\tan\theta_1 = \frac{X_C}{R_2} = \frac{10^3}{\sqrt{3}}$$

θ_1 is close to $90°$
For L-R circuit

$$X_L = \omega L = 100 \times \frac{\sqrt{3}}{10} = 10\sqrt{3}$$



$$R_1 = 10$$

$$\tan\theta_2 = \frac{X_L}{R_1}$$

$$\tan\theta_2 = \sqrt{3} \quad \Rightarrow \quad \theta_2 = \tan^{-1}(\sqrt{3})$$

$$\theta_2 = 60°$$

So, phase difference comes out $90° + 60° = 150°$
If R_2 is 20 KΩ
then phase difference comes out to be $60 + 30 = 90°$.
Therefore Ans. is **Bonus**

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Q3

(4) As $V(t) = 220 \sin 100\,\pi t$

so, $I(t) = \frac{220}{50} \sin 100\,\pi t$

i.e., $I = I_m = \sin(100\,\pi t)$

For $I = I_m$

$t_1 = \frac{\pi}{2} \times \frac{1}{100\pi} = \frac{1}{200}$ sec.

and for $I = \frac{I_m}{2}$

$\Rightarrow \frac{I_m}{2} = I_m \sin(100\,\pi t_2)$

$\Rightarrow \frac{\pi}{6} = 100\,\pi t_2 \Rightarrow t_2 = \frac{1}{600} s$

$\therefore t_{\text{req}} = \frac{1}{200} - \frac{1}{600} = \frac{2}{600} = \frac{1}{300} s = 3.3$ ms

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Q4

(4) $\omega = 100$ rad/s

We know that

$$\tan\phi = \frac{X_C}{R} = \frac{1}{\omega CR}$$

or $\tan 45^{\circ} = \frac{1}{\omega CR}$

or $\omega CR = 1$

LHS: $\omega CR = 10 \times 10 \times 10^{-6} \times 10^{3} = 1$

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